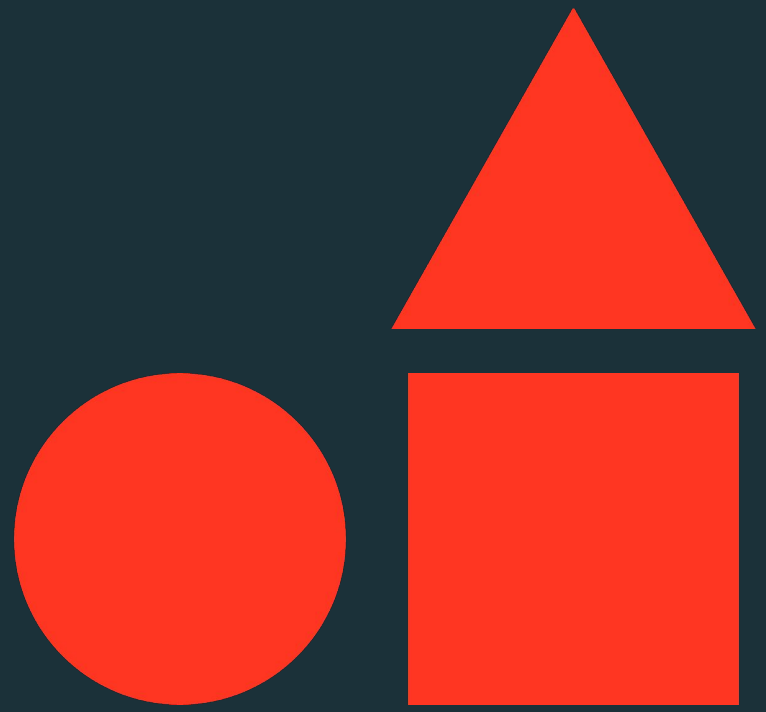
databricks

Asset Reliability Command Center

Real-time pump monitoring with governed autonomous maintenance



Data & AI World Tour
Manufacturing and energy generation

The cost of equipment downtime

Industrial benchmarks show why earlier intervention matters



$50B annually

Estimated annual cost of unplanned downtime to industrial manufacturers.



5–20% capacity

Potential reduction in a plant's productive capacity from poor maintenance strategies.



27% of electricity

Share of industrial-system electricity represented by pumps in manufacturing.

Sources: Deloitte Insights (2017); U.S. Department of Energy (2006)

Autonomous incident review

A threshold breach starts the workflow; the operator intervenes only when policy requires approval

1 Detect and assess

The App checks Lakebase each second. A vibration threshold breach creates a contextual prompt. The agent loads the governed reliability skill and estimates failure risk before the next maintenance window.

2 Prepare maintenance

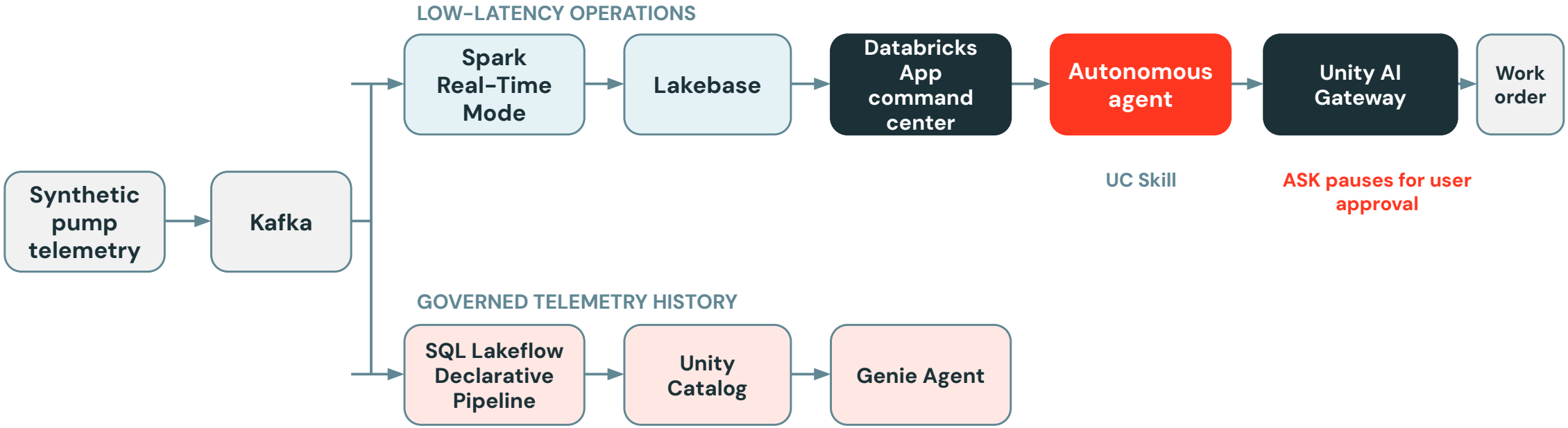
The agent discovers tools from asset_reliability_mcp, gathers evidence, and prepares a bearing replacement work order. The activity timeline exposes every model and tool event.

3 Pause for policy

Unity AI Gateway blocks the work-order call with ASK. The current user approves in context. The agent resumes, submits the order, and returns to the command center.

End-to-end Databricks architecture

One telemetry stream supports low-latency operations, governed history and autonomous action



Illustrative site value case

A 25% reduction in downtime recovers 15.0 hours and £1.2M a year

Funding executive

- Baseline downtime exposure: £4.8M
- Target downtime exposure: £3.6M
- Projected annual value: £1.2M
- Target: 25% less downtime

Reliability and operations leader

- Baseline: 60.0 downtime hours a year
- Target: 45.0 downtime hours a year
- Capacity recovered: 15.0 hours
- Two forced failures shifted to planned work

Illustrative assumptions: 24 pumps; 8 incidents × 7.5 hours = 60.0 hours; £80,000 per hour. Replace with site data.